

EDUCATION AND SCHOLASTIC ACHIEVEMENTS				
Program		Institute	CGPA	Year
B.Tech (Honors), Electrical Engineering		Indian Institute of Technology, Madras	9.30/10	2023-27
Class XII, Class X (CBSE)		National Public School HSR, Bangalore	96.6%, 96.8%	2023, 2021
- Recipient of the MV Undergraduate Prize awarded to the female student with the best all-round performance up to 6 semesters. - AIR 132 in JEE Mains 2023 out of 11.6 lakh candidates; ranked 3rd nationally among girls. AIR 1632 in JEE Advanced 2023. - AIR 14 in UG Common Entrance Exam for Design 2023 among 14,000 candidates. SAT score of 1580/1600 in 2023 (99th %ile).				
RELEVANT COURSEWORK				
Microprocessors (#) Computer Organization Quantum Comp. & Info.	Foundations of ML Adv. Topics in AI (*) Comm. Networks (#)	Foundations of Probability DSA in Python (#) Numerical Methods in C	Convex Optimization (*) Information Theory (#) Estimation Theory (*)	Digital Signal Processing (#) Communication Systems RF & Optical Communications
SKILLS				
Languages	Python, C++, C, MATLAB, AVR Assembly, ARM Assembly, Verilog, Dart, HTML, CSS, Bash, JavaScript, SQL			
ML Techniques and Libraries	Deep Learning, CNNs, RNNs, LSTMs, Transformers, Attention, Autoencoders, Time-Series Prediction, Reinforcement learning, Hyperparameter Optimization, Pytorch, TensorFlow, Keras, Scikit-learn, Optuna, NumPy, SciPy, Pandas			
Tools	Wireshark, Code Composer Studio, Git, Xilinx Vivado, LT Spice, Keil Microvision, Microchip Studio, ARM Cortex-M			
PUBLICATIONS				
Optics Express Journal, Volume 34, First Author Paper (Link) (June '26)	Attention-assisted deep neural network for nonlinear equalization & denoising in time-stretched photonic ADCs			
	- Developed a custom deep neural architecture combining attention-enhanced LSTMs and ANNs for distortion mitigation in photonic ADCs, enabling signal recovery and reducing bandwidth requirement by 3x.			
	- Presented at EE Research Scholars' Symposium at IIT Madras after being selected as a top UG research project .			
PROFESSIONAL EXPERIENCE				
Texas Instruments Intern: Embedded Software (2026)	- Summer intern in the Automotive Applications Software team for the F29x microcontroller series to satisfy marketing requirements. Worked on protecting flash memory and loading multi-core programs in single-core flash. - Demonstrated training CNNs for power grid fault detection in TI EdgeAI Studio under fixed latency & memory.			
Texas Instruments WiSH Scholar (2025)	- Selected as one of 40 among 2000+ applicants nationwide for month-long mentorship in the embedded software track ; Worked on programming PWM-controlled stepper motor systems with sensors and UART on C2000 MCUs.			
Teaching Assistant (2025)	- TA for Signals and Systems (EE1101) for 160+ students. Selected by faculty for strong performance in the course. - Conducted problem solving sessions and hands-on sessions demonstrating how to program an audio equalizer .			
RESEARCH PROJECTS AND COMPETITIONS				
RL-based Frameworks for Fiber Optical Impairment Correction (Link)	Undergraduate Research July - December 2025; Guides: Prof. LN Theagarajan, Prof. D Venkitesh			
	- Developed a single compensation system using RL frameworks to correct receiver impairments in optical channels Formulated MAB framework for polarization & dispersion compensation in 2x2 MIMO fibre-optic systems.			
	- Built UCB-based filter tap selection ; converged to optimal configuration from 5L+ options in ~1000 rounds. Extended to Q-Learning using experience replay & teacher training from classical CMA; development in progress.			
Adaptive Equalization: Optimizing DSP algorithms (Link)	Undergraduate Research January - May 2025; Guides: Prof. LN Theagarajan, Prof. D Venkitesh			
	- Implemented modifications in adaptive equalization algos. for coherent optical comm. systems achieving 8x faster convergence. Developed algo. with 95% accuracy for simultaneous compensation of dispersion and freq. offset.			
Inter-IIT Tech Meet 2025: Problem Statement by Arista Networks (Link)	Multi-Armed Bandits for Adaptive Channel Sensing and Client-Centric Wi-Fi Radio Resource Management			
	- Developed MAB algorithm for adaptive channel scheduling, replacing naive periodic scanning. 3.99% improvement in channel power estimation. Built end-to-end radio sensing pipeline combining CNN-based interference classification , MAB-based scanning, and CUSUM change detection for channel-switching.			
Dataset Condensation for Time Series Classification (Link)	- Implemented CondTSC time-series data condensation algorithm using dual-domain gradient matching in PyTorch. - Achieved 82.5% accuracy with only 4% of original training samples , comparable to full-dataset ML baselines. - Conducted cross-architecture and ablation studies, validating transferability across CNN/TCN/LSTM models .			
Makerfest: Electronics Club (2024)	- Selected in top 16 from 300+ participants in the Makerfest competition. Integrated MPU6050 and XPT2046 sensors and controllers (for touch-screen display and motion-tracking with Arduino to design a smartwatch.			
POSITIONS OF RESPONSIBILITY				
Product Design Club Head (2025)	- Led 50+ members, oversaw 9 interdisciplinary projects; Collaborated with IISc, JIPMER for projects and IITK, IITKGP, IITD for events. Secured 2 Patents under the club; led Freshie Design Challenge with 300+ participants .			
'InstiLite' Founder Product Design Club	- Founded and developed InstiLite, a full-stack mobile platform for event discovery, scheduling, and application workflows; implemented backend services with NodeJS and SQL architecture ; used Flutter for client interfaces.			
Placement Cell (2025)	- As Coordinator, planned executive framework for placement drive for 150+ students; onboarded 3 companies.			
EXTRA-CURRICULARS				
Technovation Ambassador (2021)	- India-wide ambassador for Technovation, a California-based program for school girls, addressing gender gaps in tech & entrepreneurship. Mentored 200+ girls over a year . Conducted sessions in design thinking & programming.			
Times Of India Summer Internship	- Selected as school representative to be one among 40 students chosen for a 2-week journalism internship. - Selected as guest editor by Times of India for the Newspaper In Education Guest Edition in 2020.			
Abbreviations- UG: Undergraduate, SW: Software, ADCs: Analog-to-digital converters, MAB: Multi-armed bandits, LSTM: Long short-term neural network, CNN: Convolutional Neural Network, RNNs: Recurrent Neural Network, ANNs: Artificial Neural Network. (#): Received highest grade; (*): Ongoing course;				